PABLO VILLANUEVA PICAZO

Software Engineer | AI & Data

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LinkedIn | GitHub | Portfolio

PROFILE

Software Engineer with First-Class Honours and practical experience in full-stack development. Interested in Artificial Intelligence, Machine Learning and data analytics, with academic experience in predictive modelling, data processing, automation and solutions based on Python and Edge AI.

WORK EXPERIENCE

Development Technician *April 2025 – June 2025*
Universidad Nebrija Madrid, Spain

- Full-stack development across frontend and backend, focusing on application stability and scalability.
- Implementation of REST APIs to integrate services and enable communication between systems.
- Unit and integration testing to validate software quality.

EDUCATION

BSc (Hons) Software Engineering – First-Class Honours *2025 – 2026*
York St John University York, United Kingdom

Higher Vocational Training in Multiplatform Application Development (DAM) *2023 – 2025*
Institutos Nebrija Madrid, Spain

FEATURED PROJECTS

Bitcoin Price Prediction System

- Machine Learning pipeline using Scikit-learn and Pandas to predict Bitcoin prices through technical analysis indicators.
- Edge AI architecture for decentralised real-time processing, optimising resource usage and latency.
- Final Degree Project graded 9.0/10.

Medical Records Management System (SRMS)

- Web application built with Go (net/http) and SQLite, with mitigations against OWASP Top 10 vulnerabilities.
- Security features include role-based access control (RBAC), CSRF tokens, bcrypt hashing and IP-based rate limiting.

Smart File Organizer / Local AI Daemon

- Python service with real-time file monitoring using watchdog and multi-format extraction with pypdf.
- Semantic classification using local Ollama models and structured response generation through JSON schemas.

TECHNICAL SKILLS

AI and data analytics: Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Machine Learning, time series, Edge AI and local LLMs.
Languages and frameworks: Go, Java, Python, JavaScript, TypeScript, C++, Spring Boot, React, Angular and Svelte.
Databases and cloud: SQLite, JDBC, Firebase, MySQL, PostgreSQL and cloud deployment.
Core competencies and tools: REST APIs, unit and integration testing, full-stack architecture, web security (OWASP Top 10), automation, Linux, Git, Streamlit and Ollama.

LANGUAGES AND ADDITIONAL INFORMATION

Languages: English (professional proficiency / degree completed in the UK) · Spanish (Native).
Other: Technological adaptability, continuous learning, Category B driving licence and immediate availability.